

Brain Match Game GDD

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Games Inspirations:

1. https://timmalich.github.io/edukiz_live/#/memory

1. Game Concept and Overview

- **Game Name:** Brain Match
 - **Game Genre:** Educational Puzzle / Memory / Practice
 - **Game Art Style:** Clean, Minimalist, 2D Vector Art with customizable themes.
 - **Game Brief:** A classic memory-matching game reimaged as a template for fact retention and recall. Players tap to flip cards and make pairs, actively engaging with curriculum-aligned content under various challenges. The template is highly modular and can be used to create dozens of games across subjects by simply swapping content, with each level presenting cards of increasing difficulty.
 - **Core Bloom's Level:** Remember (Recalling, Identifying, Listing)
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2. Core Gameplay Loop

2.1 Core Gameplay Loop

The game centres around a board of facedown cards with a core loop of observation, recall, and action.

- **Start Level:** A board of facedown cards is presented, along with a timer and a score counter. The card pairs are **randomly shuffled** at the start of each level to ensure replayability.
- **Tap to Reveal:** The player taps a card to reveal its content. The card briefly stays revealed.
- **Memory & Match:** The player must remember the revealed card and its location. They then tap a second card, attempting to find its match.
- **Correct Match:** If the cards match, they are removed from the board, and XP is awarded. A satisfying sound and visual effect plays.
- **Incorrect Match:** If the cards do not match, they flip back over after a short delay. A turn penalty is applied.
- **Reflex Tap Challenge:** Occasionally, a card flips over automatically, and the player must quickly tap its match before it disappears, adding a reflex-based recall element.
- **End of Level:** The level is completed once all pairs have been matched.

2.2 Educational Philosophy

- **Retrieval Practice:** The repeated act of recalling facts and their locations strengthens memory.
 - **Active Engagement:** The "tap to reveal" mechanic forces players to be active learners rather than passive consumers of information.
 - **Fact Retention:** Teaches the retention and retrieval of facts under pressure, a key skill for exams.
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3. Game Mechanics

3.1 Player Controls

- **Tap/Click to Reveal:** The primary control is a single tap or click to flip a card.

3.2 Game Board & Cards

- **Game Board:** A fixed **4x4 grid** is used across all levels.
- **Cards:** Each card has an image, a word, or a symbol on its face. Cards are created in pairs based on a defined relationship (e.g., "Chemical Symbol" and "Element Name").

3.3 Scoring & Consequences

- **Correct Match:** Cards are revealed, and XP is awarded based on the number of turns after each level.
 - **Incorrect Match:** Cards flip back over.
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4. Levels and Progression

4.1 Level Structure

The game will feature **3 distinct levels**, each with a consistent **4x4 grid**. The difficulty is increased by the complexity of the content and the scoring system.

4.2 Difficulty Scaling

- **Level 1 (Basic):** Pairs will have simple, distinct content.
- **Level 2 (Intermediate):** Pairs will be more abstract or have similar-looking content (e.g., matching a fraction to a decimal). A countdown timer is introduced.
- **Level 3 (Advanced):** Pairs will be highly abstract or visually very similar (e.g., different types of triangles), requiring deep knowledge and careful observation. The timer is shorter.

5. Educational Integration

5.1 Learning Objective

- **Objective:** To teach students to recall and identify facts, vocabulary, and formulas.

5.2 Supported Learning Formats

- **Formats:** Fact retention, vocabulary, formulas, and definitions.

5.3 Dynamic Content

- **Content Structure:** Content will be loaded via a JSON file, allowing for easy updates and creation of new topics.

6. Multiple Game Variants

This section outlines multiple game variants that can be built using the **Brain Match** template across various subjects.

1. Science: Periodic Match

- **Objective:** Match the chemical symbol to the element name.
- **Level 1:** Common elements (e.g., Fe, C, Cu, O).
- **Level 2:** More elements (e.g., He, Li, B, N, S).
- **Level 3:** Similar-looking symbols (e.g., Ca vs. Co, Mn vs. Mg).

2. English: Vocab Recall

- **Objective:** Match a word to its definition or synonym.
- **Level 1:** Simple words and clear definitions.
- **Level 2:** More complex words and nuanced definitions.
- **Level 3:** Contextual vocabulary from a specific text or topic.

3. Mathematics: Formula Match

- **Objective:** Match a formula to its name or application.
- **Level 1:** Simple formulas (e.g., "Area of a rectangle: $l \times w$ ").
- **Level 2:** More formulas (e.g., "Area of a triangle: $\frac{1}{2} \times b \times h$ ").
- **Level 3:** Advanced formulas (e.g., Pythagoras theorem).

4. Social Studies: Historic Pairs

- **Objective:** Match a historical event to its correct date or figure.
- **Level 1:** Major events (e.g., "India's Independence" to "1947").
- **Level 2:** More specific events (e.g., "Battle of Plassey" to "1757").
- **Level 3:** Figures to their achievements (e.g., "Mahatma Gandhi" to "Quit India Movement").

7. Scoring and Feedback

7.1 Scoring Metrics (Turn-Based)

The scoring is dynamic and based on the number of turns taken to complete a level.

- **Turn:** A turn is defined as flipping two cards.
- **Max XP:** The total XP available across all levels is **200 XP**.

Level	Perfect Turns (100% Accuracy)	Medium Turns	Low Turns	Max XP per Level
Level 1	12 turns (40 XP)	13-16 turns (35 XP)	16+ turns (30 XP)	40 XP
Level 2	14 turns (60 XP)	15-18 turns (50 XP)	18+ turns (40 XP)	60 XP
Level 3	16 turns (100 XP)	16-20 turns (80 XP)	20+ turns (60 XP)	100 XP

7.2 Performance Feedback

- **Metrics:** A star rating (1-3) is awarded at the end of each level based on accuracy and speed.

8. UI and Art

- **UI:** Clean and intuitive. The UI should be responsive and work well on all devices.
 - **Art:** Simple, minimalist vector art that can be easily customised with new colours or icons for each game variant.
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9. NCERT Variants

This section outlines how the **Brain Match** template can be applied to specific topics in the NCERT curriculum.

9.1 Science

- **Topic: Force** (Class 6)
 - **Pairs:** Match a type of force to its example (e.g., "Friction" to "Rubbing hands together").
- **Topic: Living & Non-living Things** (Class 6)
 - **Pairs:** Match a picture of an object to its classification (e.g., a picture of a car to "Non-living," a picture of a dog to "Living").

9.2 Mathematics

- **Topic: Fractions & Decimals** (Class 7)
 - **Pairs:** Match a fraction to its decimal equivalent (e.g., "1/4" to "0.25").
- **Topic: Basic Geometry** (Class 6)
 - **Pairs:** Match a picture of a shape to its name (e.g., an equilateral triangle to "Equilateral Triangle").

9.3 Social Science

- **Topic: Historical Events** (Class 6)
 - **Pairs:** Match a historical event to its correct date (e.g., "India's Independence" to "1947").
- **Topic: Government** (Class 8)
 - **Pairs:** Match a term to its definition (e.g., "Democracy" to "Government by the people").

9.4 English

- **Topic: Vocabulary & Grammar** (Class 7)
 - **Pairs:** Match a word to its synonym, antonym, or correct definition.
- **Topic: Parts of Speech** (Class 6)
 - **Pairs:** Match a word to its part of speech (e.g., "Quickly" to "Adverb").

10. References/ Sources

- <https://www.mathsisfun.com/games/math-match-game.html>
- <https://pbskids.org/daniel/games/barnyard-match/>

Game Variants

Variant 1: Science: The Living World Match

This variant focuses on classifying living things and their properties.

- **Level 1: Basic Classification** (8 pairs, 4x4 grid)
 - **Objective:** Match an animal to its category.
 - **Pairs:** Dog ↔ Mammal, Sparrow ↔ Bird, Frog ↔ Amphibian, Fish ↔ Aquatic Animal, Rose ↔ Plant, Butterfly ↔ Insect, Snake ↔ Reptile, Whale ↔ Mammal (to introduce a nuance).
 - Image Assets of the first pair, like Dog, Sparrow, etc
 - And the text will be of Mammal, Bird, etc.....
 - **Level 2: Plant Parts & Functions** (8 pairs, 4x4 grid)
 - **Objective:** Match a plant part to its function.
 - **Pairs:** Leaves ↔ Photosynthesis, Roots ↔ Anchoring, Stem ↔ Support, Flower ↔ Reproduction, Seed ↔ Spreading, Stomata ↔ Gas Exchange, Chlorophyll ↔ Green Pigment, Pollen ↔ Fertilisation.
 - Image Assets of the first pair, like Leaves, Roots, etc
 - And Text will be of Photosynthesis, Anchoring, etc.....
(updated)
Leaves ↔ Food
 - Roots ↔ Grip
 - Stem ↔ Hold
 - Flower ↔ Seed
 - Seed ↔ Grow
 - Stoma ↔ Air
 - Green (for Chlorophyll) ↔ Color
 - Pollen ↔ Dust
 - **Level 3: Ecosystem & Habitat** (8 pairs, 4x4 grid)
 - **Objective:** Match an animal to its specific habitat.
 - **Pairs:** Camel ↔ Desert, Penguin ↔ Polar Region, Fish ↔ Water, Tiger ↔ Forest, Polar Bear ↔ Arctic, Kangaroo ↔ Grassland, Camelion ↔ Forest, Octopus ↔ Ocean Floor.
 - Image Assets of the first pair, like Camel, Penguin, etc
 - And the Text will be of Desert, Polar, etc.....
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Variant 2: Human Body Match

This variant focuses on matching parts of the human body to their functions or locations.

Body Assets Images will be there, and the function will be text.

- **Objective:** Match an organ to its primary function.
- **Level 1:** Match major organs to their simple functions.
 - **Pairs:** Brain ↔ Thinking, Heart ↔ Pumping Blood, Lungs ↔ Breathing, Stomach ↔ Digesting Food, Kidneys ↔ Filtering Blood, Bones ↔ Supporting the Body, Muscles ↔ Movement, Liver ↔ Detoxification.
- **Level 2:** Match organs to a more specific function or location.
 - **Pairs:** Arteries ↔ Carry blood away from the heart, Veins ↔ Carry blood to the heart, Nervous System ↔ Controls the body, Skeletal System ↔ Provides a framework, Villi ↔ Absorb nutrients, Cornea ↔ Focusing light, Pancreas ↔ Produces insulin, Oesophagus ↔ Connects throat to stomach.
- **Level 3:** Match small-scale body parts to their function.
 - **Pairs:** Neurons ↔ Transmit nerve impulses, Alveoli ↔ Air sacs in lungs, Nephrons ↔ Filtering units in kidneys, Capillaries ↔ Tiny blood vessels, Rods and Cones ↔ Detect light in eyes, Tendons ↔ Connect muscle to bone, Ligaments ↔ Connect bone to bone, Platelets ↔ Help with clotting.

Game Variant 3: Periodic Match

This variant focuses on matching chemical symbols to their corresponding element names. Text will be there for both.

Level 1: Common Elements

- **Objective:** Match the chemical symbol of a common element to its name.
- **Pairs:** Fe ↔ Iron, C ↔ Carbon, Cu ↔ Copper, O ↔ Oxygen, Al ↔ Aluminium, S ↔ Sulphur, Au ↔ Gold, Cl ↔ Chlorine.

Level 2: Intermediate Elements

- **Objective:** Match a wider range of elements, including some less common ones.
- **Pairs:** He ↔ Helium, Li ↔ Lithium, B ↔ Boron, N ↔ Nitrogen, S ↔ Sulphur, P ↔ Phosphorus, Zn ↔ Zinc, I ↔ Iodine.

Level 3: Advanced Elements

- **Objective:** Match elements with similar-looking symbols or names, requiring a high level of recall.
 - **Pairs:** Ca ↔ Calcium, Co ↔ Cobalt, Mn ↔ Manganese, Mg ↔ Magnesium, K ↔ Potassium, Na ↔ Sodium, Pb ↔ Lead, Br ↔ Bromine.
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Variant 4: Shapes Match

The shape images will be there, and the shape name will be text.

Level 1: Basic Shapes

Pairs: (Image: a square) ↔ Square, (Image: a triangle) ↔ Triangle, (Image: a circle) ↔ Circle, (Image: a rectangle) ↔ Rectangle, (Image: a star) ↔ Star, (Image: a rhombus) ↔ Rhombus, (Image: a pentagon) ↔ Pentagon, (Image: a hexagon) ↔ Hexagon

Level 2: Advanced 2D Shapes

Pairs: (Image: a right-angled triangle) ↔ Right-angled Triangle, (Image: a parallelogram) ↔ Parallelogram, (Image: an octagon) ↔ Octagon, (Image: an oval) ↔ Oval, (Image: a trapezoid) ↔ Trapezoid, (Image: a kite) ↔ Kite, (Image: a crescent) ↔ Crescent, (Image: a semicircle) ↔ Semicircle

Level 3: Introduction to 3D Shapes

Pairs: (Image: a sphere) ↔ Sphere, (Image: a cone) ↔ Cone, (Image: a cube) ↔ Cube, (Image: a pyramid) ↔ Pyramid, (Image: a cylinder) ↔ Cylinder, (Image: a rectangular prism) ↔ Rectangular Prism, (Image: a triangular prism) ↔ Triangular Prism, (Image: a torus) ↔ Torus

Variant 5: Opposites Match

This variant focuses on matching a word to its correct antonym. The difficulty increases by introducing more subtle, complex, or formal words.

Both will be text only.

Level 1: Simple Antonyms

- **Pairs:** Big ↔ Small, Hot ↔ Cold, Happy ↔ Sad, Up ↔ Down, Fast ↔ Slow, Open ↔ Close, Day ↔ Night, Old ↔ New

Level 2: Intermediate Antonyms

- **Pairs:** Ancient ↔ Modern, Shallow ↔ Deep, Simple ↔ Complex, Arrive ↔ Depart, Weak ↔ Strong, Awake ↔ Asleep, Dull ↔ Sharp, Tidy ↔ Messy

Level 3: Advanced Antonyms

- **Pairs:** Transparent ↔ Opaque, Abundant ↔ Scarce, Artificial ↔ Authentic, Crucial ↔ Trivial, Hostile ↔ Friendly, Flexible ↔ Rigid, Innocent ↔ Guilty, Generous ↔ Selfish

Variant 6: Flag Match

This variant focuses on matching flags to their corresponding countries or regions. The Flag Image will be there, and the country name will be text.

Level 1: Iconic Flags

- **Pairs:** (Image of the Indian flag) ↔ India, (Image of the American flag) ↔ USA, (Image of the Japanese flag) ↔ Japan, (Image of the Brazilian flag) ↔ Brazil, (Image of the French flag) ↔ France, (Image of the Chinese flag) ↔ China, (Image of the Australian flag) ↔ Australia, (Image of the British flag) ↔ United Kingdom

Level 2: Continental Flags

- **Pairs:** (Image of the Egyptian flag) ↔ Egypt, (Image of the Nigerian flag) ↔ Nigeria, (Image of the Kenyan flag) ↔ Kenya, (Image of the Ethiopian flag) ↔ Ethiopia, (Image of the South African flag) ↔ South Africa, (Image of the Moroccan flag) ↔ Morocco, (Image of the Ghanaian flag) ↔ Ghana, (Image of the Algerian flag) ↔ Algeria

Level 3: World Flags

- **Pairs:** (Image of the Polish flag) ↔ Poland, (Image of the Indonesian flag) ↔ Indonesia, (Image of the Slovakian flag) ↔ Slovakia, (Image of the Slovenian flag) ↔ Slovenia, (Image of the Irish flag) ↔ Ireland, (Image of the Côte d'Ivoire flag) ↔ Côte d'Ivoire, (Image of the Nigerian flag) ↔ Nigeria, (Image of the Ghanaian flag) ↔ Ghana

Level 1: Iconic Flags

- India
 - USA
 - Japan
 - Brazil
 - France
 - China
 - Mexico (*replaces Australia* ✗)
 - U.K (*replaces United Kingdom* ✗)
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Level 2: Continental Flags

- Egypt
 - Kenya (*replaces Nigeria* ✗)
 - Ghana (*replaces Ethiopia* ✗)
 - Morocco
 - Sweden
 - Algeria
 - Sudan (*replaces South Africa* ✗)
 - Nepal (*extra short country added for balance*)
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Level 3: World Flags

- Poland
- Ireland
- Ghana (*replaces Indonesia* ✗)
- Togo (*replaces Slovakia* ✗)
- Benin (*replaces Slovenia* ✗)
- Mali (*replaces Côte d'Ivoire* ✗)
- Niger (*replaces Nigeria* ✗)
- Laos (*kept as a short filler to keep balance*)

(1st) Grade 1st Brainmatch Variants

Grade 1 Mathematics Variants

Chapter 2: What is Long? What is Round?

Game Name: Shape Friends

Level 1: Similar Shape Objects – Basic

Format:

Card A = image of object

Card B = image of another object with a similar shape

Image Card (Object)	Image Card (Similar Shape Object)
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Football	Beach Ball
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Matchbox	Book
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Birthday Cap	Paper Cone
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Glass	Water Bottle
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Level 2: Similar Shape Objects – Familiar Objects

Format:

Card A = image of object

Card B = image of another object with a similar shape

Image Card (Object)	Image Card (Similar Shape Object)
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Ball	Orange
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Matchbox	Pencil Box
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Birthday Cap	Funnel
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Glass	Bottle
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Level 3: Similar Shape Objects – Different Looking Objects

Format:

Card A = image of object

Card B = image of another object with a similar overall shape

Image Card (Object)	Image Card (Similar Shape Object)
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Watermelon	Ball
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Shoe Box	Book
----------	------

Ice-Cream Cone	Birthday Cap
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Tumbler	Water Bottle
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Level 4: Similar Shape Objects – Mixed Challenge

Format:

Card A = image of object

Card B = image of another object with a similar shape

Image Card (Object)	Image Card (Similar Shape Object)
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Orange	Football
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Notebook	Matchbox
----------	----------

Funnel	Paper Cone
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Tumbler	Bottle
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For Grade 1, all these cards should remain **image ↔ image**, with no text required on the cards.

Chapter 3: Number Play / Counting & Number Representation

Game Name: Number Friends

Core Learning Objective:

Understand that the **same number can be represented in different visual ways** — as a numeral, fingers, dots, birds, fruits, or other groups of objects.

Core Brain Match Relationship:

Numeral ↔ Equivalent Quantity / Representation

For Grade 1, keep the game **highly visual**. Text should be avoided apart from the numerals themselves.

Level 1: Numeral ↔ Dots — Numbers 1–4

Grid: 2×4 — 4 pairs

Format:

Card A = numeral

Card B = image showing the same number of dots

Numeral Card	Quantity Image Card
--------------	---------------------

1	•
2	••
3	•••
4	••••

Difficulty: Very Easy

This establishes the basic rule: **the numeral represents how many objects are shown.**

Level 2: Numeral ↔ Fingers — Numbers 2–5

Grid: 2×4 — 4 pairs

Format:

Card A = numeral

Card B = hand image showing the same number of fingers

Numeral Card	Finger Image Card
2	Hand showing 2 fingers
3	Hand showing 3 fingers
4	Hand showing 4 fingers
5	Hand showing 5 fingers

Difficulty: Easy

This introduces a different representation without increasing the board size.

Level 3: Fingers ↔ Dots — Numbers 1–4

Grid: 2×4 — 4 pairs

Format:

Card A = hand image

Card B = dots showing the same quantity

Finger Image Card	Dot Image Card
Hand showing 1	1 dot
Hand showing 2	2 dots
Hand showing 3	3 dots
Hand showing 4	4 dots

Difficulty: Easy–Medium

This is important because the child can no longer depend on a written numeral. They must recognise that **two different visual representations mean the same quantity**.

Level 4: Numeral ↔ Object Groups — Numbers 3–6

Grid: 2×4 — 4 pairs

Format:

Card A = numeral

Card B = image containing the same number of familiar objects

Numeral Card	Object Quantity Image
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3	3 Apples
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4	4 Birds
---	---------

5	5 Flowers
---	-----------

6	6 Balls
---	---------

Difficulty: Medium

Use **large, clearly separated objects** so Grade 1 students can count them quickly.

Level 5: Numeral ↔ Object Groups — Numbers 5–8

Grid: 4×4 — 8 pairs

Format:

Card A = numeral

Card B = visual quantity

Numeral Card	Quantity Image Card
---------------------	----------------------------

2	2 Apples
---	----------

3	3 Dots
---	--------

4 4 Fingers

6 6 Balls

Difficulty: Medium

This is the first **2×4 board**.

Even though different representation types appear, the relationship remains completely consistent:

Numeral ↔ same quantity

Level 6: Mixed Number Representations — Numbers 2–9

Grid: 2×4 — 4 pairs

Format:

Card A = numeral

Card B = mixed visual representation of that quantity

Numeral Card	Matching Representation
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2	Hand showing 2
---	----------------

4	4 Dots
---	--------

6	6 Apples
---	----------

7	7 Stars
---	---------

Difficulty: Medium–Hard

This becomes the main mastery level because children cannot predict whether the matching card will contain **fingers, dots, or objects**.

Level 7: Visual ↔ Visual Number Match

Grid: 2×4 — 4 pairs

Format:

Card A = one visual representation

Card B = a completely different visual representation of the same quantity

Visual Card A	Visual Card B
2 Fingers	2 Dots
4 Fingers	4 Apples
6 Birds	6 Balls
8 Flowers	8 Dots

Difficulty: Hard

There are **no numerals as hints** here.

The child must understand:

Different-looking groups can still represent the same number.

This is the strongest conceptual Brain Match level for the chapter.

Difficulty Progression

Level 1: Numeral ↔ simple dots

→ **Level 2:** Numeral ↔ fingers

→ **Level 3:** Fingers ↔ dots

→ **Level 4:** Numeral ↔ real objects

→ **Level 5:** Larger 4×4 board

→ **Level 6:** Mixed representations

→ **Level 7:** Visual ↔ visual with no numeral support

Chapter 12: How Much Can We Spend?Game Name: Money Match

Core Learning Objective:

Help children recognise Indian currency, find the total value of small groups of coins/notes, and understand that **different combinations of money can represent the same amount**.

The chapter introduces ₹1, ₹2, ₹5 and ₹10 coins and ₹10, ₹20, ₹50 and ₹100 notes, then progresses into finding totals and making the same amount in different ways.

Level 1: Coin ↔ Denomination

Grid: 2×4 — 4 pairs

Format:

Card A = image of coin

Card B = denomination text

Currency Image Card	Denomination Card
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₹1 Coin	₹1
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₹2 Coin	₹2
₹5 Coin	₹5
₹10 Coin	₹10

Difficulty: Very Easy

This is pure currency recognition.

Level 2: Note ↔ Denomination

Grid: 2×4 — 4 pairs

Format:

Card A = image of currency note

Card B = denomination text

Currency Image Card	Denomination Card
₹10 Note	₹10
₹20 Note	₹20
₹50 Note	₹50
₹100 Note	₹100

Difficulty: Easy

Keep the note images visually large enough for Grade 1 students to distinguish them easily.

Level 3: Money Combination ↔ Total Amount

Grid: 2×4 — 4 pairs

Format:

Card A = image of coins/notes together

Card B = total amount

Money Combination Card Total Card

₹5 + ₹1 ₹6

₹5 + ₹2 ₹7

₹5 + ₹2 + ₹1 ₹8

₹5 + ₹5 ₹10

Difficulty: Easy–Medium

The child now moves from simply recognising currency to **adding small amounts**.

The chapter explicitly asks children to find total amounts from groups of coins and notes.

Level 4: Money Combination ↔ Total Amount — Mixed

Grid: 2×4 — 4 pairs

Format:

Card A = coin/note combination

Card B = total monetary value

Money Combination Card Total Card

₹2 + ₹1 ₹3

₹2 + ₹2 ₹4

₹5 + ₹1 ₹6

₹10 + ₹1

₹11

Difficulty: Medium

This is the first 2×4 level.

Each total is intentionally unique so that every card has **only one valid match**.

Level 5: Same Amount ↔ Same Amount — Introduction

Grid: 2×4 — 4 pairs

Format:

Card A = one currency combination

Card B = a different combination representing exactly the same amount

Combination A	Combination B
₹2 + ₹1	₹1 + ₹1 + ₹1
₹2 + ₹2	₹2 + ₹1 + ₹1
₹5 + ₹1	₹2 + ₹2 + ₹2
₹5 + ₹2	₹5 + ₹1 + ₹1

Difficulty: Medium

There are **no total-number cards** here.

The child has to recognise:

Different groups of money can have the same value.

This directly follows the chapter's instruction to form amounts using different currency combinations.

Level 6: Same Amount ↔ Same Amount — Master Challenge

Grid: 2×4 — 4 pairs

Format:

Card A = one currency combination

Card B = a different currency combination with equal value

Combination A	Combination B	Actual Value
₹2 + ₹1	₹1 + ₹1 + ₹1	₹3
₹2 + ₹2	₹2 + ₹1 + ₹1	₹4
₹5	₹2 + ₹2 + ₹1	₹5
		₹6
		₹7
		₹8
		₹10
₹10 + ₹2	₹5 + ₹5 + ₹2	₹12

Difficulty: Hard

Important: The **Actual Value** column is only for the developer/GDD. It should **not appear on the player's cards**.

This is the strongest Brain Match level because both sides look completely different, yet represent exactly the same value.

The textbook itself ends with “**Match the same amount by drawing a line**”, showing different groups of coins/notes that represent equal amounts.

Difficulty Progression

Level 1: Coin Image ↔ Denomination

→ **Level 2:** Note Image ↔ Denomination

→ **Level 3:** Simple Combination ↔ Total

→ **Level 4:** Mixed Combination ↔ Total on 4×4

→ **Level 5:** Combination ↔ Different Equal Combination

→ **Level 6:** 4×4 Equal-Value Mastery

Total Recommended Levels: 6

The most important Brain Match mechanic is ultimately:

$$\text{₹10} + \text{₹2} \leftrightarrow \text{₹5} + \text{₹5} + \text{₹2}$$

The child isn't matching pictures because they look alike — they're matching them because they have the **same mathematical value**. That's exactly why this is one of the strongest Brain Match chapters in Grade 1.

Grade 1 English Variant

Chapter 1: Two Little Hands

Game Name: Body & Sound Match

Core Learning Objective:

Build recognition of **body-part vocabulary**, **body-part functions**, and **beginning letter sounds** through simple visual matching.

The chapter explicitly asks children to **match body parts with their pictures**, teaches functions such as *eyes* → *see*, *ears* → *hear*, *nose* → *smell*, *mouth* → *eat*, and later introduces initial sounds from **A–Z** using familiar objects and pictures.

Level 1: Body Picture ↔ Body Word

Grid: 2×4 — 4 pairs

Format:

Card A = image of body part

Card B = body-part word

Image Card	Word Card
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Eye image	EYE
-----------	-----

Ear image	EAR
-----------	-----

Nose image	NOSE
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Hand image	HAND
------------	------

Difficulty: Very Easy

This directly mirrors the textbook's picture-to-body-part matching activity.

Level 2: More Body Picture ↔ Body Word

Grid: 2×4 — 4 pairs

Format:

Card A = image of body part

Card B = body-part word

Image Card	Word Card
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Mouth image	MOUTH
-------------	-------

Head image	HEAD
------------	------

Leg image	LEG
-----------	-----

Foot image	FOOT
------------	------

Difficulty: Easy

This expands the child's body vocabulary without changing the interaction.

Level 3: Body Part ↔ Function

Grid: 2×4 — 4 pairs

Format:

Card A = image of body part

Card B = short action/function word

Body Part Image	Function Card
------------------------	----------------------

Eyes	SEE
------	-----

Ears	HEAR
------	------

Nose	SMELL
------	-------

Mouth	EAT
-------	-----

Difficulty: Easy–Medium

These relationships are taught directly in the chapter's reading activity.

Level 4: Body Part ↔ Action

Grid: 2×4 — 4 pairs

Format:

Card A = image of body part

Card B = short action word

Body Part Image	Action Card
Hands	CLAP
Legs	WALK
Head	MOVE
Skin	FEEL

Difficulty: Medium

The poem and activities explicitly connect hands with clapping, legs with walking, head with moving and skin with feeling.

Letter Sound Section

For these levels:

Card A = Letter

Card B = image of an object beginning with that sound

The **object name does not need to appear on the gameplay card**. Ideally, use only the object image so the child connects the letter with its beginning sound.

Level 5: Beginning Sounds A–D

Grid: 2×4 — 4 pairs

Letter Card	Picture Card
A	Ant
B	Bag

C	Cat
---	-----

D	Dog
---	-----

Difficulty: Easy

Level 6: Beginning Sounds E–H

Grid: 2×4 — 4 pairs

Letter Card	Picture Card
--------------------	---------------------

E	Egg
---	-----

F	Fish
---	------

G	Goat
---	------

H	Hut
---	-----

Difficulty: Easy

Level 7: Beginning Sounds I–P

Grid: 2×4 — 4 pairs

Letter Card	Picture Card
--------------------	---------------------

I	Ice-cube
---	----------

M	Mango
---	-------

N	Nose
---	------

P	Parrot
---	--------

Difficulty: Medium

This is a good point to move from a **2×4 board to a 4×4 board** because children already understand the matching mechanic.

The chapter explicitly introduces these phonics associations.

Level 8: Beginning Sounds Q–X

Grid: 2×4 — 4 pairs

Letter Card	Picture Card
-------------	--------------

R	Rat
---	-----

S	Sun
---	-----

T	Table
---	-------

U	Umbrella
---	----------

Difficulty: Medium–Hard

Level 9: Final Sounds + Mixed Revision

Grid: 2×4 — 8 pairs

Letter Card	Picture Card
-------------	--------------

Z	Zebra
---	-------

B	Bee
---	-----

C	Cap
---	-----

E	Ear
---	-----

Difficulty: Hard

The extra A–E examples are also supported by the chapter's dedicated **Letter Sounds** activity, which provides multiple words for the same initial letter—for example **a** → **axe/ant/arm**, **b** → **bag/bee/boy**, and **c** → **cart/cap/cat**.

This makes the final level useful rather than simply repeating the exact same pictures.

Recommended Progression

Level 1: Picture ↔ Body Word

→ **Level 2:** More Body Vocabulary

→ **Level 3:** Body Part ↔ Function

→ **Level 4:** Body Part ↔ Action

→ **Level 5:** A–D Initial Sounds

→ **Level 6:** E–H Initial Sounds

→ **Level 7:** I–P on 4×4

→ **Level 8:** Q–X on 4×4

→ **Level 9:** Y–Z + Mixed Phonics Revision

Total Recommended Levels: 9

This is one chapter where **9 levels are justified**, because there are really two substantial Brain Match learning strands:

Body vocabulary/functions + initial letter sounds.

They shouldn't be mixed together within a single board, but they can comfortably exist as different levels within **Body & Sound Match**.

Chapter: Picture Time / Five Little Monkeys

Game Name: Animal Friends

Core Learning Objective:

Help children recognise common animals and associate an **animal picture with its English name**.

The chapter ends with an explicit “**Match the animals with their names**” activity using Lion, Monkey, Fish, Elephant, Frog and Rabbit. Earlier, the picture discussion also explicitly asks children about **birds and snakes**, giving us enough chapter-supported vocabulary for a later 4×4 board.

Level 1: Animal Picture ↔ Animal Name

Grid: 2×4 — 4 pairs

Format:

Card A = animal image

Card B = animal name

Image Card	Word Card
------------	-----------

Lion image	LION
------------	------

Monkey image	MONKEY
--------------	--------

Fish image	FISH
------------	------

Frog image	FROG
------------	------

Difficulty: Easy

Keep the animal illustrations large and visually distinct.

Level 2: Animal Picture ↔ Animal Name — Full Match

Grid: 2×4 — 4 pairs

Format:

Card A = animal image

Card B = animal name

Image Card	Word Card
Lion image	LION
Monkey image	MONKEY
Fish image	FISH
Frog image	FROG

Difficulty: Medium

The first six animals come directly from the chapter's matching exercise, while **snake** and **bird** are explicitly referenced in the opening picture discussion.

Grade 2nd Brianmatch Game Variants

Grade 2 Mathematics BrainMatch Game Variants

Chapter: 2- Shapes Around Us (Already Built)

BrainMatch (Shapes Match) Game: https://sanwiksagarpw.github.io/BrainMatch_Shapes/

Chapter 4: Shadow Story (Togalu)

Game Name: Shadow Match

Core Learning Objective:

Help children understand that an object or hand pose produces a corresponding **shadow/silhouette**, and recognise objects from their shadow shapes.

The opening activity explicitly shows children making different animal shadows using their hands, while the next activity asks children to shine a torch on objects and observe the shapes of their shadows. The textbook visually demonstrates a **cube, cone, cylinder and ball with their shadows**, which gives us a very clean Brain Match interaction.

Core Relationship:

Object / Hand Pose Image ↔ Matching Shadow Image

Card Style:

Image ↔ Image only.

No text is needed on the cards.

Level 1: Easy Object Shadows

Grid: 2×4 — 4 pairs

Format:

Card A = coloured object image

Card B = black/grey silhouette produced by that object

Object Image Card

Shadow Card

Cube	Cube shadow
Cone	Cone shadow
Cylinder	Cylinder shadow
Ball	Ball shadow

Difficulty: Easy

These four objects and their shadows are directly illustrated in the chapter's **“Fun with Shadows”** activity.

The silhouettes are visually different, so this is a strong introductory board.

Level 2: Hand Pose ↔ Animal Shadow

Grid: 2×4 — 4 pairs

Format:

Card A = image of hands/fingers making an animal pose

Card B = the corresponding animal-shaped shadow

Hand Pose Card	Matching Shadow Card
Hand Pose A	Matching Animal Shadow A
Hand Pose B	Matching Animal Shadow B
Hand Pose C	Matching Animal Shadow C
Hand Pose D	Matching Animal Shadow D

Difficulty: Medium

For implementation, these should use the **four hand-pose + shadow examples shown on the opening page of the chapter**, rather than introducing unrelated animals.

The player sees the actual hands on one card and has to remember which silhouette those hands create.

This is harder than Level 1 because:

The hand itself may look very different from the final animal silhouette.

Level 3: Shadow Challenge

Grid: 4×4 — 8 pairs

Format:

Card A = source image

Card B = matching shadow

Source Image Card	Shadow Card
Cube	Cube shadow
Cone	Cone shadow
Cylinder	Cylinder shadow
Ball	Ball shadow
Hand Pose A	Animal Shadow A
Hand Pose B	Animal Shadow B
Hand Pose C	Animal Shadow C
Hand Pose D	Animal Shadow D

Difficulty: Hard

This level combines the **four object-shadow examples** with the **four hand-shadow examples** already established in the chapter.

So we actually do have **8 curriculum-supported pairs**, meaning a 4×4 final board is justified here without inventing extra content.

Difficulty Progression

Level 1: Object ↔ obvious object shadow — **2×4**

→ **Level 2:** Hand pose ↔ animal silhouette — **2×4**

→ **Level 3:** Mixed objects + hand shadows — **4×4**

Total Recommended Levels: 3

This is a good case where the 4×4 level works naturally because we're not stretching the chapter just to fill the board.

Chapter 9: Which Season is it?

Game Name: Clock Match

Core Learning Objective:

Help children read **whole-hour time on an analog clock** and connect the position of the hour/minute hands with the correct numerical time.

The chapter's “**Gargi's Day**” activity presents eight analog clocks across a child's daily routine, and the next activity explicitly asks students to **read the time on clocks** and draw hands for given times.

Core Brain Match Relationship:

Analog Clock Image ↔ Digital Time

Card Format:

Card A = analog clock image

Card B = short numerical time such as 7:00

No activity names or sentences should appear on the cards.

Level 1: First Clock Match

Grid: 2×4 — 4 pairs

Format:

Card A = analog clock

Card B = matching time

Clock Image Card	Time Card
Clock showing 6 o'clock	6:00
Clock showing 7 o'clock	7:00
Clock showing 8 o'clock	8:00
Clock showing 9 o'clock	9:00

Difficulty: Easy

All clocks show exact hours, so the minute hand remains at **12** and the child mainly learns to identify the hour hand.

Level 2: More Clock Times

Grid: 2×4 — 4 pairs

Format:

Card A = analog clock

Card B = matching time

Clock Image Card	Time Card
Clock showing 1 o'clock	1:00
Clock showing 2 o'clock	2:00
Clock showing 4 o'clock	4:00
Clock showing 10 o'clock	10:00

Difficulty: Medium

These times come from the remaining clocks shown in **Gargi's Day**: coming home, lunch, playing with friends and studying.

Level 3: Full Clock Challenge

Grid: 4×4 — 8 pairs

Format:

Card A = analog clock

Card B = matching numerical time

Clock Image Card	Time Card
Clock showing 1 o'clock	1:00
Clock showing 2 o'clock	2:00
Clock showing 4 o'clock	4:00
Clock showing 6 o'clock	6:00
Clock showing 7 o'clock	7:00
Clock showing 8 o'clock	8:00
Clock showing 9 o'clock	9:00

Clock showing 10 o'clock 10:00

Difficulty: Hard

This uses **all eight unique whole-hour clocks shown in Gargi's daily schedule**, so the 4×4 board is fully justified rather than being artificially filled.

Level Progression

Level 1: 4 familiar clock times → **2×4**

Level 2: 4 additional clock times → **2×4**

Level 3: all 8 times mixed → **4×4**

Total Levels: 3

Important Implementation Rule

Keep the relationship extremely clean:



Analog clock ↔ 7:00

Not:



↔ **“Gargi draws water at 7 o’clock.”**

Chapter 10 — Fun at the Fair

Game Name: Money Match (Same as Grade 1st)

Grade 2 English Variant

Chapter 3: It is Fun

Game Name: Animal Action Match

Core Learning Objective:

Help children associate familiar animals with the **specific action words taught in the chapter**, while reinforcing vocabulary such as **hop, swim, trot, fly, leap, climb, crawl, and swing**.

The poem directly teaches relationships such as **lamb → leap, cat → climb, frog → hop, fish → swim, horse → trot, bird → fly**, and the later matching activity adds **snake → crawl, squirrel → run, monkey → swing**.

Core Brain Match Relationship:

Animal Image ↔ One Short Action Word

No sentences on cards.

Level 1: Easy Animal Actions

Grid: 2×4 — 4 pairs

Format:

Card A = animal image

Card B = one action word

Animal Image Card	Action Card
Frog	HOP
Fish	SWIM
Bird	FLY
Cat	CLIMB

Difficulty: Easy

These four actions are visually and conceptually very different, making them a strong introduction for Grade 2.

The poem explicitly teaches **hop like a frog, swim like a fish, fly like a bird, and climb like a cat.**

Level 2: More Animal Actions

Grid: 2×4 — 4 pairs

Format:

Card A = animal image

Card B = one action word

Animal Image Card	Action Card
-------------------	-------------

Horse	TROT
-------	------

Lamb	LEAP
------	------

Snake	CRAWL
-------	-------

Monkey	SWING
--------	-------

Difficulty: Medium

The horse and lamb relationships are introduced in the poem, while the later “**Match the animals to their actions**” activity explicitly reinforces horse, lamb, snake, squirrel and monkey action relationships.

Level 3: Animal Action Challenge

Grid: 4×4 — 8 pairs

Format:

Card A = animal image

Card B = one action word

Animal Image Card	Action Card
-------------------	-------------

Frog	HOP
------	-----

Fish	SWIM
------	------

Bird	FLY
------	-----

Cat	CLIMB
Horse	TROT
Lamb	LEAP
Snake	CRAWL
Monkey	SWING

Difficulty: Hard

This combines all eight previously learned relationships into one 4×4 memory board.

The difficulty now comes from:

remembering 8 different animal-action relationships

rather than increasing reading complexity.

Level Progression

Level 1: 4 highly familiar actions → **2×4**

→ **Level 2:** 4 less familiar/new action words → **2×4**

→ **Level 3:** all 8 mixed together → **4×4**

Total Levels: 3

Card Design Rule

Keep the cards exactly this short:

 ↔ HOP
 ↔ SWIM
 ↔ TROT
 ↔ CRAWL

Do **not** use:

“The frog hops.”

“The horse trots.”

Grade 3rd Brainmatch Variants

Grade 3 Science Variants

Chapter 4: Getting to Know Plants

Level 1: Plant Part ↔ Function

Format:

Card A = image of plant part

Card B = short function text

Image Card (Plant Part)	Text Card (Function)
-------------------------	----------------------

Roots	Absorb Water
Stem	Support Plant
Leaf	Make Food
Flower	Make Fruit
Fruit	Protect Seeds
Seed	New Plant
Bark	Outer Cover
Trunk	Hold Branches

Level 2: Plant Type ↔ Example

Format:

Card A = image of plant example

Card B = plant type text

Image Card (Plant Example)	Text Card (Plant Type)
----------------------------	------------------------

Mango	Tree
Rose	Shrub
Tulsi	Shrub
Mint	Herb
Tomato	Herb
Wheat	Grass
Money Plant	Climber
Pumpkin	Creeper

Grade 4th Brainmatch

Science

Chapter 7: How Things Work

Topic: Float / Sink and Spin / Roll

Why it fits strongly:

Objects can be classified by behaviour or property.

Card A: Object Image	Card B: Behaviour Text
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Stone	Sinks
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Leaf	Floats
------	--------

Ball	Rolls
------	-------

Top	Spins
-----	-------

Cork	Floats
------	--------

Coin	Sinks
------	-------

Bangle	Spins
--------	-------

Empty Bottle	Floats
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